7 Sewerage and Sewage Treatment Implications

7.1 Introduction

7.1.1.1 The proposed Yuen Long South (YLS) development scheme will generate large amount of sewage flows due to the population and employment intake. The expected impact / implication on the existing and planned sewerage systems and the proposals of sewerage infrastructure and mitigation measures to support YLS development are discussed in this chapter.

7.2 Legislation, Standards and Guidelines

7.2.1.1 The following established principles and guidelines are adopted for sewage flow estimation, assessment and evaluation of sewerage and sewage treatment implications of the Project:

- EPD's Technical Paper Guidelines for Estimating Sewage Flows (GESF) for Sewerage Infrastructure Planning Version 1.0;
- DSD's Sewerage Manual Part 1 (Key Planning Issues and Gravity Collection System) and Part 2 (Pumping Stations and Rising Mains), Third Edition, May 2013;
- Hong Kong Planning Standards and Guidelines (HKPSG); and
- EIAO-TM Annex 14.

7.3 Methodology and Design Criteria

7.3.1 Objectives and Procedures

7.3.1.1 The objectives and procedures of the sewerage impact assessment are summarized as follows:

- To conduct a desktop review to assess the available capacities in the existing sewerage facilities, including committed works, within and in the vicinity of the Study Area;
- To derive the sewage flow and load projection arising from the developments in the Study Area and the connected sewerage catchment;
- To examine the potential impact (if any) arising from new (additional) sewage generation from the proposed development on the existing sewerage infrastructure; and
- To identify, evaluate and recommend the necessary mitigation measures and an appropriate sewage collection, treatment and disposal proposal for the PDA.

7.3.1.2 The assessment has been carried out in accordance with the guidelines set out in EPD's Technical Paper Guidelines for Estimating Sewage Flows (GESF) for Sewerage Infrastructure Planning Version 1.0 and DSD's Sewerage Manual.

7.3.2 Unit Flow Factors

7.3.2.1 The following unit flow factors in **Table 7.1** extracted from the GESF are adopted:

Table 7.1 Unit Flow Factors for Different Land Uses

Land Use as specified in GESF	Unit Flow Factor (m^3/person/day)	Remarks
Residential – Public, Private (R1)	0.19(1)	Equivalent to RSc, R1(SSF)c, R2(SSF)c & R1c in RODP
Residential – Private (R2)	0.27(1)	Equivalent to R2c in RODP
Residential – Private (R3), (R4)	0.37(1)	Equivalent to R3, R4, R5, R6 & OU(MU) in RODP
Schools (E)	0.04(2) for Students 0.08(2) for Staff	-
Government, Institution or Community (G/IC)	0.28(2)	Equivalent to G, OU(RCP&SPS), OU(ESS), OU(STW) in RODP; Sum of flows due to employee (0.08m^3/person/day) & J11 related activities (0.2m^3/person/day)
Commercial (C)	0.28(2)	Equivalent to R1(SSF)c, R2(SSF)c, R2c & OU(MU) in RODP; Sum of flows due to employee (0.08m^3/person/day) & J4 related commercial activities (0.2m^3/person/day)
Industrial and Open Storage (I/OS)	0.18(2)	Equivalent to OU(SU)1, OU(SU)2, OU(S+W) & OU(OS) in RODP
General Permanent Housing – Yuen Long, Mui Wo (EDA)	0.25(1)	The retained existing developments embraces mixed residential settlement including existing residential clusters, two existing non-indigenous villages, three existing religious institutions and committed residential developments

Notes: (1) Referenced from EPD GESF Table T-1
(2) Referenced from EPD GESF Table T-2

7.3.3 Peaking Factors

7.3.3.1 Peaking factors cater for seasonal/diurnal fluctuation and normal amount of infiltration and inflow. The peaking factors shall be adopted in accordance with EPD's GESF as shown in **Table 7.2**.

7.3.3.2 Under normal condition, peaking factors (excluding stormwater allowance) are applicable to planning sewerage facilities receiving flow from new upstream sewerage systems which essentially have no misconnections and defects for infiltration. If the service conditions of the upstream sewerage systems for the planning horizons under

consideration are unclear, peaking factors (including stormwater allowance) shall be used. For design purpose, the peaking factors (including stormwater allowance) will be used.

Table 7.2 Peaking Factors for Various Population Ranges

Population Range	Peaking Factor (including stormwater allowance) for facility with existing upstream sewerage	Peaking Factor (excluding stormwater allowance) for facility with new upstream sewerage
Sewers		
< 1,000	8	6
1,000 – 5,000	6	5
5,000 – 10,000	5	4
10,000 – 50,000	4	3
> 50,000	Max ($7.3/N^{0.15}$, 2.4)	Max ($6/N^{0.175}$, 1.6)
Sewage Treatment Works, Preliminary Treatment Works and Pumping Stations		
< 10,000	4	3
10,000 – 25,000	3.5	2.5
25,000 – 50,000	3	2
> 50,000	Max ($3.9/N^{0.065}$, 2.4)	Max ($2.6/N^{0.065}$, 1.6)

Notes:

Referenced from EPD GESF Table T-5

N = Contributing population in thousands.

7.3.4 Unit Load Factors

7.3.4.1 The global unit load factors used to estimate the sewage loading from the proposed developments are referred in Table 4 of the Sewerage Manual Part 1 as listed in **Table 7.3**.

Table 7.3 Unit Load Factors for Different Land Uses

Trades	Unit	SS (kg/d)	BOD (kg/d)	COD (kg/d)	TKN (kg/d)	NH_3-N (kg/d)	*E. coli* (no./d)
Domestic (Residential)	Person	0.04	0.042	0.09	$8.5x10^{-3}$	$5 x10^{-3}$	$4.3x10^{10}$
Commercial, G/IC and Industrial (1)	Employee	0.059	0.087	0.173	$9.2x10^{-3}$	$4.8 x10^{-3}$	$3.5x10^{10}$
Schools (2)	Student	0.034	0.034	0.07	$6.7x10^{-3}$	$4 x10^{-3}$	$3.5x10^{10}$

Notes:

Referenced from Sewerage Manual Part 1 Table 4

(1) Unit Load Factors (Commercial, G/IC and Industrial) is the sum of Unit Load Factors for commercial activities and employed population.

(2) Unit Load Factors of Schools is the Unit Load Factor of employed population.

7.3.5 Interfacing Projects

7.3.5.1 There are a number of planned, on-going and completed sewerage improvement works and sewage treatment projects within and in the vicinity of the Study Area, as summarized in **Table 7.4**.

Table 7.4 Interfacing Projects

Item No. / Project Ref.	Name of Project	Actual or Programmed Completion Date
CE37/2012(DS)	Improvement of Yuen Long Town Nullah (Town Centre Section) – Stage 1 Improvement Works – D&C	2026 (Programmed Completion Date)
DC/2013/10 (PWP Item No.4411DS)	"Design, Build and Operate San Wai Sewage Treatment Works – Phase 1"	Sept 2020 (Programmed Completion Date)

7.4 Existing and Planned Sewage Infrastructure

7.4.1 Existing Sewerage System

7.4.1.1 The YLS PDA is located within the sewerage catchment of San Wai Sewage Treatment Works (SWSTW). The existing sewage generated within the catchment including the PDA is currently conveyed sequentially to Ha Tsuen Sewage Pumping Station (HTSPS) and SWSTW for treatment. The effluent from SWSTW is then transported to the Northwest New Territories effluent tunnel and the Urmston Road submarine outfall for deep discharge to the North Western Water Control Zone (WCZ).

7.4.1.2 As per DSD's Agreement No. CE 43/2007 (DS) "Yuen Long and Kam Tin sewerage treatment upgrade – Upgrading of San Wai Sewage Treatment Works - Investigation, Design and Construction", HTSPS and SWSTW have existing capacities of 164,000m^3/day (ADWF), while the capacity of the Urmston Road submarine outfall is 295,000m^3/day. With further reference to DSD's Contract No. DC/2013/10, namely "Design, Build and Operate San Wai Sewage Treatment Works – Phase 1" which has sought EIA approval, the treatment capacity of the existing SWSTW is being upgraded to 200,000m^3 per day and being enhanced its sewage treatment level to incorporate chemical treatment with disinfection.

7.4.1.3 At close proximity to YLS PDA is the sewerage catchment of Yuen Long Sewerage Treatment Works (YLSTW), covering the urban area of Yuen Long and Kam Tin which provides sewage treatment before discharge to Deep Bay in the north.

7.4.1.4 The YLSTW is a secondary treatment works with a design capacity of 70,000m^3/day (ADWF) and a peaking factor of 3.

7.4.1.5 With reference to DSD's drainage record plans, the existing sewerage systems in and around the PDA is shown on **Figure 7.2** and described below:

- For Tong Yan San Tsuen (TYST), an existing public sewerage system collects sewage generated in TYST to SWSTW. Sewage flows from the western part of TYST is conveyed in an east to west direction towards SWSTW via HTSPS, while the sewage flows from the eastern portion is conveyed via the

Ping Shun Street SPS located immediately south of the West Rail Long Ping Station.

- For the eastern side of the PDA along Kung Um Road, there is an intermediate Lung Tin Sewage Pumping Station (LTSPS) situated at the junction of Yuen Long Highway and Kung Um Road, transferring the sewage collected from (i) the existing sewers along Sham Chung Road and (ii) the planned sewers constructed under PWP Item No. 368DS to the sewerage network leading to HTSPS and eventually to SWSTW. Currently, there is no existing public sewerage system within eastern side of the PDA along Kung Um Road and the sewage generated from the existing villages and open storage areas within these parts of the PDA are believed to be generally disposed of by means of private sewerage system (i.e. septic tanks and soakaway systems) or expedient discharge to the Yuen Long Main Nullah.

7.4.1.6 A Yuen Long South trunk sewerage network under PWP Item No. 368DS and DSD's Contract No. DC/2009/08 "Construction of Yuen Long South Branch Sewers and Expansion of Ha Tsuen Sewage Pumping Station" was completed in Mar 2013, which comprises of 4 nos. of DN300-DN450 branch sewers along Kiu Hing Road, Shui Tsiu San Tsuen Road, Tai Tong Road and Tai Shu Ha Road East and a new SPS at Shui Tsiu San Tsuen Road as shown in **Figure 7.2**. It aims to discharge the sewage collected from unsewered areas in Shap Pat Heung to the existing DN900 trunk sewer along Sham Chung Road and subsequently towards SWSTW for treatment and disposal.

7.4.2 Planned Sewerage System

Yuen Long South Branch Sewerage

7.4.2.1 A planned Yuen Long South branch sewerage system to collect the sewage generated from Shap Pat Heung and Tai Tong Tsuen was being constructed for completion in 2016 under PWP Item Nos. 384DS and DSD's Contract No. DC/2012/05. The works comprise the construction of gravity sewers for 6 unsewered areas, namely Nam Pin Wai, Sai Pin Wai, Tsoi Uk Tsuen, Wong Uk Tsuen, Ying Lung Wai and Tai Tong Tsuen, among which the branch sewer in Tai Tong Tsuen as shown in **Figure 7.2** was connecting to the planned Yuen Long South trunk sewer constructed under Contract No. DC/2009/08 mentioned above.

Upgrading of San Wai Sewage Treatment Works

7.4.2.2 According to DSD's Contract No. DC/2013/10 "Design, Build and Operate San Wai Sewage Treatment Works – Phase 1" and the information from EPD, the upgrading of SWSTW will increase its treatment capacity to an ADWF of 200,000m^3/day in 2021 to meet the future population growth in the sewerage catchment and improve the effluent quality through promoting its treatment level to chemically enhanced primary treatment (CEPT) with ultraviolet disinfection. It is understood from EPD that the current proposal for upgrading SWSTW has not considered the YLS PDA recommended under this Study.

7.4.2.3 Although the Hung Shui Kiu New Development Area (HSK NDA) are located within the sewerage catchment of SWSTW which is being upgraded, according to information provided from CEDD, a new local STW in HSK NDA will be provided to handle sewage generated from the NDA itself.

Yuen Long Effluent Polishing Plant

7.4.2.4 DSD is implementing an upgrading works to increase the treatment capacity of YLSTW from 70 000 m^3/day to 150 000 m^3/day while to concurrently upgrade the YLSTW to an effluent polishing plant with tertiary treatment level under CE 3/2015 (DS), Yuen Long Effluent Polishing Plant – Investigation, Design and Construction. The project has been commenced in May 2016 and the anticipated date of completion is Nov 2028.

Improvement Works of Yuen Long Town Nullah (Town Centre Section)

7.4.2.5 DSD is implementing the Rehabilitation of Yuen Long Nullah (Town Centre Section) project under CE 37/2012 (DS), Improvement of Yuen Long Town Nullah (Town Centre Section) – Stage 1 Improvement Works – Design and Construction. These include the construction of a dry weather flow (DWF) interception system, scenic water supply system and beautification works at the town centre section of Yuen Long Main Nullah. The dry weather flow interception system, which will construct new DWF interceptors along Yuen Long Nullah to convey flows downstream to YLSTW. The entire project is anticipated for completion in 2020.

Other Planned Development with associated Sewerage Works

7.4.2.6 There are also numerous concurrent development projects within the vicinity of the Project which involved associated sewerage works for discharging their sewage to respective STW for treatment and disposal. These ongoing developments include:

- CEDD's Engineering Works at Wang Chau, Yuen Long (Formerly, HKHA's Public Housing Development at Wang Chau);
- PlanD's Housing Development at Kam Tin South MTRC West Rail Kam Sheung Road Station and Pat Heung Maintenance Depot;
- Various development by private developers;
- Sewage demands arising from existing demands of Yuen Long STW Catchment, planned growth through the year of population intake;
- Sewage demands arising from the provision of public sewerage for unsewered villages in Yuen Long areas; and
- Sewage demands arising from New Territories North Development (i.e. Lok Ma Chau PDA and Ngau Tam Mei PDA and San Tin area).
- CEDD and PlanD's HSK NDA;
- Housing Development in Tan Kwai Tsuen, Lam Tei; and
- Various development by private developers.

7.5 Estimation of Sewage Flows and Loads

7.5.1 Sewage Flows Estimation

Sewage Flows Generation from YLS PDA

7.5.1.1 The population intake for YLS will commence from 2027 until 2038 with total population of about 88,000 people and employment of 10,500 people.

7.5.1.2 As per the methodology and technical approach described in **Section 7.3**, the sewage flows generated from the YLS PDA at **2038** are estimated to be **about 23,635 m^3/day** (ADWF) which is summarised in **Table 7.5**.

Table 7.5 Sewage Estimation for YLS PDA

Land Use Type	Parameter	Remarks
Residential		
Public Housing (RS, R1(SSF), R2(SSF))		
Population	53,100	From Proposed Development Schedule
Unit Flow Factor (m^3/person/day)	0.190	Table T-1, EPD's GESF
ADWF (m^3/day)	**10,089**	
Private Housing (R1c, R2, R2c, R3, R4, R5, R6, OU(MU))		
Population (R1c)	2,100	From Proposed Development Schedule
Unit Flow Factor (m^3/person/day)	0.190	Table T-1, EPD's GESF
ADWF (m^3/day)	**399**	
Population (R2, R2c)	15,900	From Proposed Development Schedule
Unit Flow Factor (m^3/person/day)	0.270	Table T-1, EPD's GESF
ADWF (m^3/day)	**4,293**	
Population (R3, R4, R5, R6, OU(MU))	14,400	From Proposed Development Schedule
Unit Flow Factor (m^3/person/day)	0.370	Table T-1, EPD's GESF
ADWF (m^3/day)	**5,328**	
Population (EDA)	2340	Existing Residential Population
Unit Flow Factor (m^3/person/day)	0.250	Table T-1, EPD's GESF
ADWF (m^3/day)	**585**	
Education - Schools (E)		
No. of Schools	8	Primary Schools
Students per School	765	Assuming each school has 30 classrooms with each classroom 25.5 persons. Table 4, Chapter 3, HKPSG
No. of Students	6,120	
Unit Flow Factor (m^3/person/day)	0.04	Table T-2, EPD's GESF
No. of Staff	504	From Key Development Parameters
Unit Flow Factor (m^3/person/day)	0.08	Table T-2, EPD's GESF
ADWF (m^3/day)	**285**	
Government/Institution Community		
Community, Social Services (J11)		
No. of Employees	744	From Key Development Parameters
Unit Flow Factor (m^3/employee/day)	0.280	J11, Table T-2, EPD's GESF

Land Use Type	Parameter	Remarks
ADWF (m^3/day)	**208**	
Public Utilities (J2)		
No. of Employees	160	From Key Development Parameters
Unit Flow Factor (m^3/employee/day)	0.330	J2, Table T-2, EPD's GESF
ADWF (m^3/day)	**53**	
Commercial		
No. of Employees	7,640	From Key Development Parameters
Unit Flow Factor (m^3/employee/day)	0.280	J4, Table T-2, EPD's GESF
ADWF (m^3/day)	**2,139**	
Storage and Workshop + Open Storage		
No. of Employees	1,421	From Key Development Parameters
Unit Flow Factor (m^3/employee/day)	0.180	J3, Table T-2, EPD's GESF
ADWF (m^3/day)	**256**	
Summary		
Total ADWF (m^3/day)	**23,635**	

Notes:
ADWF = Development Parameter x Unit Flow Factor

7.5.2 Pollutant Loads Estimation

7.5.2.1 As per the methodology in **Section 7.3** and the unit load factors in **Table 7.3**, estimation of pollutant loads arising from the proposed PDA is detailed in **Table 7.6**.

Table 7.6 Projected Pollutant Loads for YLS PDA

Trades	Development Parameter	SS (kg/d)	BOD (kg/d)	COD (kg/d)	TKN (kg/d)	NH_3-N (kg/d)	*E. coli* (no./d)
Domestic (Residential)	87,840(1)	3,514	3,689	7,906	747	439	37.8×10^{14}
Commercial, G/IC & Industrial	9965(2)	588	867	1,724	92	48	3.5×10^{14}
Schools	6624(3)	225	225	464	44	26	2.3×10^{14}
Total		**4,327**	**4,781**	**10,093**	**883**	**513**	**44×10^{14}**

Notes:
(1) YLS PDA total residential population
(2) YLS PDA total employment
(3) YLS PDA total students and staff
Pollutant Load = Development Parameter x Unit Load Factor (Table 7.3)

7.5.2.2 With the total projected **ADWF of 23,635m^3/day** and the projected pollutant loads as shown in **Table 7.6**, the average sewage concentration is estimated and shown on **Table 7.7**.

Table 7.7 Projected Pollutant Concentrations

Pollutant Concentration					
SS (mg/L)	**BOD (mg/L)**	**COD (mg/L)**	**TKN (mg/L)**	**NH_3-N (mg/L)**	***E. coli* (no./100ml)**
183.1	**202.3**	**427.1**	**37.4**	**21.7**	**1.84 x10^7**

Note:
Pollutant Concentration = Pollutant Load (Table 7.6) ÷ Total ADWF

7.6 Potential Impacts to Sewerage systems and Recommended Mitigation Measures

7.6.1 Constraints of Existing Sewerage Catchments

7.6.1.1 Yuen Long, Kam Tin and HSK areas are undergoing rapid developments. Both the SWSTW and YLSTW have capacity constraints to handle future catchment flows. Major upgrading works to expand treatment capacities and upgrade treatment levels of both the SWSTW and YLSTW are being implemented. However, none of the above upgrading sewage treatment facilities can cater the proposed development of the YLS PDA.

7.6.1.2 To provide protection to the inland and marine water quality of the Deep Bay WCZ, there is a stringent requirement for "No Net Increase in Pollution Loading", which requires the developments within the Deep Bay catchment areas do not result in an increase in pollution loads to the inland and marine waters. Sewage from the sewerage catchment of YLSTW are being discharged to the Deep Bay WCZ after treatment. If the sewage flows from the YLS PDA are proposed to be discharged to Deep Bay, this requirement will need to be complied with. Therefore, potential mitigation measures and offsetting of pollution credits would be required.

7.6.2 Impacts to Existing and Planned Sewerage Systems

Existing Sewage Treatment Works

7.6.2.1 As described in **Section 7.4**, the YLS PDA is located within the SWSTW sewerage catchment, while at close proximity to the adjoining sewerage catchment of YLSTW. Due to the presence of other on-going major developments within these 2 sewerage catchments, the sewage flow arising from these developments shall be considered collectively.

7.6.2.2 EPD has advised that both the upgraded capacity of YLSTW (ultimately 150,000m^3/day ADWF) and SWSTW (200,000m^3/day ADWF) are inadequate to accommodate the flows from their respective sewerage catchments in **2031** and **2034** respectively. To accommodate the additional flow arising from YLS, a new sewerage infrastructure (including STW, SPS, sewers and rising mains) will be required as presented in **Section 7.6.3**.

Existing Lung Tin Sewage Pumping Station

7.6.2.3 The existing Lung Tin Sewage Pumping Station (LTSPS) at eastern side of the PDA along Kung Um Road, which transfers the sewage collected from (i) the existing DN900 sewers along Sham Chung Road and (ii) the planned sewers constructed under PWP Item No. 368DS to the sewerage network leading to HTSPS and eventually SWSTW has been considered in the development of the RODP. As shown in **Figure 7.1**, the existing LTSPS will be retained in its present location and unaffected by the proposed YLS PDA.

Existing Sewers at Tong Yan San Tsuen

7.6.2.4 The existing areas at TYST are served by an existing public sewerage system of sewer sizes between 225mm and 600mm. To suit the new development layout at TYST, these existing sewers would be replaced by the proposed sewerage system of YLS.

7.6.3 Recommended Sewage Treatment and Disposal Strategy

Sewage Treatment by On-site Sewage Treatment Works to Tertiary Level

7.6.3.1 Following numerous meetings, liaison and coordination amongst CEDD, PlanD, EPD and DSD, it is proposed that the sewage generated from the YLS PDA will be discharged to a new local STW, namely YLS STW for on-site treatment, which is located to the southern end of the YLS PDA (see **Figure 7.1**). The on-site STW will be designed to achieve a treatment capacity of 24,000m^3/day in ADWF and up to tertiary level treatment standard to enable for reuse of treated sewage effluent (TSE) as reclaimed water while the rest of sewage effluent is suggested to be treated by secondary plus treatment (with UV disinfection and 75% nitrogen removal) and discharged to the Urmston Road submarine outfall.

Reuse of TSE as Reclaimed Water Reuse in YLS PDA, Existing Area and Adjacent Developments

7.6.3.2 The new STW will provide tertiary sewage treatment to produce TSE of high quality for conversion into reclaimed water that is suitable for reuse within the PDA and adjacent developments for non-potable uses such as toilet flushing.

7.6.3.3 Moreover, as part of the urban design of the YLS PDA, the existing Yuen Long Nullah (with some sections to be partially decked to meet the traffic demand) adjacent to the eastern side of the PDA along Kung Um Road will be revitalised by application of both soft and hard landscaping treatments in order to improve its aesthetics and to promote a water friendly culture within the YLS PDA. A small amount of TSE discharged to Yuen Long Nullah can be regarded as a continuous scenic water source as part of the waterscape provided within the revitalised nullah.

7.6.3.4 However, as the TSE generated would be excessive that it cannot be totally reused by YLS PDA itself, the tertiary TSE should be properly considered in view of the no net increase in the pollution loading requirement of the Deep Bay WCZ.

7.6.3.5 As there are also some adjacent developments, such as Long Bin, Tan Kwai Tsuen and Tin Shui Wai, liaison with the relevant project proponents and government departments regarding adopting reclaimed water for non-potable uses (mainly for toilet flushing) are being conducted.

Pollution Credit Offsetting by Existing Livestock Farms

7.6.3.6 There are currently 6 number of existing pig and chicken farms scattered in YLS PDA and it is proposed that 5 of these livestock farms would be removed. As it is understood that AFCD are implementing Livestock Keeping License for operation of these livestock farms, the farm operator/ owners shall treat their livestock waste to meet minimum discharge standards as stipulated in the Waste Disposal (Livestock Waste) Regulations (Chapter 354A).

7.6.3.7 Hence the removal of the pollution loadings generated from these livestock farms are identified as one of the distinct source contributing to the pollution credit in Deep Bay. With some of these existing livestock farms being removed by YLS, their associated pollution credit can hence be released and offset by equivalent amount of pollution loading being discharged from YLS STW to Yuen Long Nullah.

Further Polishing of TSE by Reedbed

7.6.3.8 In the previous project namely Planning and Engineering Study on Development of Lok Ma Chau Loop and existing Tin Shui Wai Wetland Park, it has been proven that reedbed is capable to further polish the TSE by means of biological breakdown process. There are approximate 4 ha areas adjacent to YLS STW in the RODP as shown in **Figure 7.1**, which would be developed into a reedbed. A small amount of TSE from YLS STW will pass through the reedbed for further polishing before discharging into Yuen Long Nullah.

7.6.3.9 However, due to the uncertain pollution removal efficiency of the reedbed, its polishing function would only be regarded as an extra provision and would not be considered in the pollution load offset balance as shown in the following sections and **Appendix 7.1**.

Surplus TSE Export Disposal Strategy

7.6.3.10 Having made reference to tertiary TSE disposal strategy in HSK NDA, the surplus TSE will be discharged from YLS STW to existing inlet chamber of the NWNT effluent tunnel at San Wai for discharging to the Urmston Road submarine outfall for deep discharge to the North Western WCZ.

7.6.3.11 With reference to the Environmental Impact Assessment report approved in 2003 for Upgrading and Expansion of SWSTW, effluent discharge of 246,000 m^3/day from the upgraded SWSTW (Phase 1) equipped with Chemical-enhanced Preliminary Treatment (CEPT) with disinfection, are within an acceptable level. Further study on approved EP-464/2013 "*Upgrading and expansion of San Wai Sewage Treatment Works and Expansion of Ha Tsuen Pumping Station*" and approved EIAR-203/2016 "*Hung Shui Kiu New Development Area*", total quantities of effluent discharge via the Urmston Road submarine outfall can be summarised in **Table 7.8**.

Table 7.8 Summary of Effluent Discharge from HSK NDA and SWSTW – Phase 1

SWSTW – Phase 1 with CEPT (m^3 per day)	Total TSE from HSK NDA to be discharged (with Secondary Plus Treatment)	Discharge to the Urmston Road Submarine Outfall (m^3 per day)
200,000	45,696	200,000 (CEPT) + 45,696 (Secondary Plus)

Note: Secondary Plus Treatment is a combination of secondary treatment with 75% nitrogen removal and UV disinfection.

7.6.3.12 Further studies have been carried out on pollution loading of combined effluents and hydraulic capacity assessment of the Urmston Road submarine outfall and the

NWNT effluent tunnel. The summarised results are shown in **Table 7.9** and **Table 7.10**. It can be seen that pollution loading from combined effluent of SWSTW, HSK STW and YLS STW are within the EIA requirements and flow rate of combined effluent are within the hydraulic capacity of the NWNT effluent tunnel and the Urmston Road submarine outfall. **Figure 7.5** shows the proposed sewer alignment from YLS STW to inlet chamber of the NWNT effluent tunnel at San Wai for discharging to the Urmston Road submarine outfall.

Table 7.9 Summary of Pollution Loading of Combined Effluents

Parameter	Unit	SWSTW (CEPT) (1)	HSK STW (Secondary+)(2)	YLS STW (Secondary+)	Total Combined Effluent	EIA No. 072/2003
E. Coli	cfu/day	4.00E+12	4.57E+10	1.00E+10	4.06E+12	4.92E+12
Total residual chlorine	kg/day	-	-	-	-	-
Dissolved oxygen	kg/day	-	-	-		-
TSS	kg/day	11,000	1,371	300	12,671	13,530
Colour	HU	-	-	-	-	-
Turbidity	NTU	-	-	-	-	-
pH	-	-	-	-	-	-
Threshold odour number	TON	-	-	-	-	-
BOD5	kg/day	20,000	914	200	21,114	24,600
Ammonia nitrogen	kg/day	5,000	91	20	5,111	6,150
Total Inorganic Nitrogen (TIN)	kg/day	5,000	366	80	5,446	6,150
Synthetic detergents	kg/day	-	-	-	-	-
Temperature	degC	-	-	-	-	-
Chloride	kg/day	-	-	-	-	-

Table 7.10 Summary of Hydraulic Capability

	Effluent (m^3/day)
SWSTW (CEPT)(1)	200,000
HSK STW (Secondary+)(2)	45,696
YLS STW (Secondary+)	10,000
Total Combined Effluent	255,696
Capacity of the NWNT Effluent Tunnel	400,000
Capacity of the Urmston Road Submarine Outfall	295,000

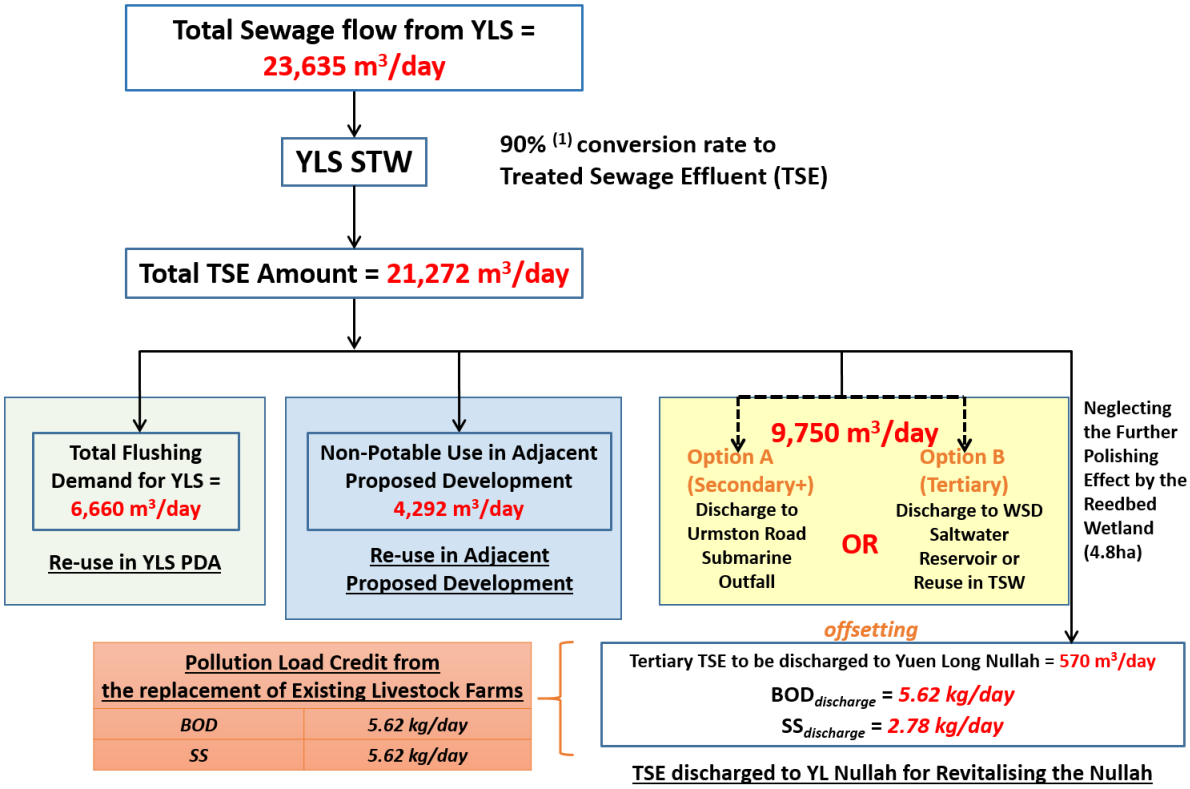
Note:
(1) Ref: Appendix 5A, approved EIA report for the Upgrading and Expansion of San Wai STW and the Expansion of Ha Tsuen PS.
(2) Ref: Appendix 5.2, approved EIA report of HSK NDA

7.6.3.13 Alternatively, WSD is implementing a saltwater supply system in the existing NWNT area including Tin Shui Wai and Yuen Long Town area. It can be considered to extend the use of surplus reclaimed water as a supply source to the new saltwater supply system. Liaison with WSD and Department of Health (DH) on transferring surplus reclaimed water to Tan Kwai Tsuen (TKT) saltwater service reservoir for being utilised by its supply zone is being conducted concurrently.

Overall Sewage Treatment and Disposal Strategy

7.6.3.14 As shown in the schematic **Diagram 1**, the generated TSE volume from YLS STW would be 21,272m^3/day by assuming 90% conversion rate from the sewage, taking into account of the sludge volume, system loss from the treatment process and transmission loss in the distribution system.

Diagram 1 – Schematic Diagram of the overall sewage treatment and disposal strategy



Note: (1) With reference of EIA Report for Hung Shui Kiu New Development Area Planning and Engineering Study – Investigation, minimum 90% of treated sewage can be reused.

7.6.3.15 As per the latest development parameter, there will be 6,660m^3/day of water demand for toilet flushing in YLS that could be supplied by reclaimed water (water demand calculation for flushing toilet in YLS PDA can be referred to **Appendix 7.2**). For other potential reclaimed water use in YLS such as landscape irrigation, water make-up for water features and etc., as their water demand are subject to seasonal variation and acceptance of other government departments, they will not be considered in the overall reclaimed water reuse plan at this stage until further details of these infrastructures and acceptance by the corresponding government has been sought.

7.6.3.16 As advised by the project proponents of Long Bin and Tan Kwai Tsuen Development, their flushing water demand were estimated to about 4,292m^3/day (2,375m^3/day from Long Bin Development and 1,917m^3/day from Tan Kwai Tsuen Development) in total. As such, part of the reclaimed water, i.e. 4,292m^3/day could be discharged to these development for their flushing uses.

7.6.3.17 As indicated in the schematic diagram above, a small amount of TSE of about 570 m^3/day could be discharged into Yuen Long Nullah. This amount of flow was calculated using in a conservative approach, thus the overall pollution to Deep Bay is anticipated to be less than existing situation. The pollution loading of 570m^3/day of TSE being discharged could be offset by the pollution credit gained from those

existing livestock farms (including 3 pig farms and 2 chicken farms) removed due to the development of YLS PDA, which comply with the requirement for "No Net Increase in Pollution Loading" into Deep Bay. For more details including the quantification of pollution credit from existing livestock farms and TSE being discharged can refer to **Appendix 7.1**. On the contrary, if surplus TSE is not proposed to be discharged to Yuen Long Nullah after removal of living livestock farms, it is considered that it will result in an improvement to the water quality of the downstream Deep Water Bay. Liaison with EPD on further review the actual pollution loading generated from these livestock farms and the exact of TSE to be discharged to Yuen Long Nullah in compliance with the requirement of "No Net Increase in Pollution Loading" into Deep Bay is being conducted.

7.6.3.18 As mentioned in **Section 7.6.3.10,** the surplus TSE can be conveyed to the existing inlet chamber of the NWNT effluent tunnel at San Wai for discharging to the Urmston Road submarine outfall at North Western WCZ. The surplus TSE is suggested to be treated by secondary plus treatment (with UV disinfection and 75% nitrogen removal).

7.6.3.19 Alternatively, liaison with WSD and DH is being conducted such that 9,750m^3/day of reclaimed water would be transferred to TKT saltwater service reservoir for being utilised by its supply zone or modifying WSD existing saltwater supply system in existing Tin Shui Wai area to replace saltwater by reclaimed water for flushing use. However, DH expressed their concern on potential associated human health risks, as they noted that the existing salt water supply system also supplies to some existing fire-hydrants in its supply zone, which could potentially be another route of human exposure to reclaimed water. Further negotiation is ongoing with WSD and DH concurrently.

7.6.4 Proposed Sewerage System for YLS Development

7.6.4.1 A new public sewerage network will be proposed to convey sewage flow from various lots within YLS PDA to the proposed on-site STW for treatment and disposal. **Figure 7.3** shows the preliminary layout of the proposed sewerage system for the YLS PDA. The majority of alignment of the proposed trunk sewerage network is either along the proposed roads or through amenities area. Drainage reserves will be required for laying sewer pipes outside the roads such as through amenities areas or open spaces.

7.6.4.2 Gravity sewerage system will be provided as far as possible. To overcome topographical constraints and to avoid excessive sewer depth, two new pumping stations (SPSs) are proposed to convey the sewage flows to the proposed YLS STW. The locations and catchments of the proposed SPSs are shown in **Figure 7.3**.

7.6.4.3 **Table 7.11** to **Table 7.13** show the accumulated projected sewage flow, peaking factors and accumulated projected peak flows respectively for SPS1 and SPS2 for Stage 1 to Stage 4. The breakdown calculation of sewage flow can be referred to **Appendix 7.3**.

Table 7.11 Estimated Sewage Flows For Stage 1 to Stage 4

Facilities	Accumulated Sewage Flow (m^3/day)			
	Stage 1	Stage 2	Stage 3	Stage 4
SPS1	597	900	900	4,754
SPS2	2,534	8,643	19,773	23,627*

Note:

* Small part of catchment will directly flow to STW by gravity pipe and thus the accumulated sewage flow would not be identical to the total sewage flow as shown in Diagram 1

Table 7.12 Peaking Factors for Proposed SPS

Facilities	Peaking Factors (PF)			
	Stage 1	Stage 2	Stage 3	Stage 4
SPS1	4.0	4.0	4.0	3.5
SPS2	4.0	3.0	2.4	2.4

Table 7.13 Estimated Peak Flows for Stage 1 to Stage 4

Facilities	Accumulated Projected Peak Sewage Flow (l/s)			
	Stage 1	Stage 2	Stage 3	Stage 4
SPS1	28	42	42	193
SPS2	117	300	549	656

7.6.4.4 A network of water mains and a service reservoir will be constructed for the reuse of TSE as reclaimed water for non-potable uses such as toilet flushing within YLS PDA and adjacent developments. The surplus TSE treated by secondary plus treatment can be conveyed to the existing inlet chamber of the NWNT effluent tunnel at San Wai for discharging to the Urmston Road submarine outfall at North Western WCZ. Alternatively, all reclaimed water could be treated up to the proposed reuse quality and then be transferred to TKT saltwater service reservoir for being utilised by its supply zone, which is under negotiation with WSD and DH.

7.6.4.5 Estimated volume and target water quality standard of reclaimed water is summarised in **Table 7.14** below.

Table 7.14 Estimated Volume of Reclaimed Water and Disposal of TSE

Description	Volume (m^3/day)
Total sewage flow	23,635
Total TSE Amount*	21,272
Reuse of TSE as reclaimed water and Discharging to Yuen Long Nullah (Tertiary Treatment)	11,522
Surplus TSE to be discharged to the Urmston Road submarine outfall (Secondary Plus Treatment)	9,750

Note:

* With reference of EIA Report for Hung Shui Kiu New Development Area Planning and Engineering Study – Investigation, minimum 90% of treated sewage can be reused

7.6.4.6 The treatment of reclaimed water up to the proposed reuse quality presented in **Table 7.15** will be provided within the 4-ha footprint of YLS STW, including the chlorine contact tank, chemical storage, reclaimed water storage and distribution pumps connecting to the supply pipe network, as detailed in **Figure 7.4**. Adjacent to the YLS STW, there is an adjoining reedbed for further polishing of a small amount of TSE before discharging to Yuen Long Nullah as mentioned in **Section 7.6.3**.

Table 7.15 Proposed Water Quality Standards of Reclaimed Water Reuse

Water Quality Parameter	Unit	Recommended Standard for Flushing, Landscape Irrigation and DCS Make-up
E. Coli	cfu/100mL	Not detectable
Total residual chlorine (TRC)	mg/L	>1 (out of treatment system) >0.2 (at point-of-use)

Water Quality Parameter	Unit	Recommended Standard for Flushing, Landscape Irrigation and DCS Make-up
Dissolved oxygen (DO)	mg/L	>2
TSS	mg/L	<5
Colour	HU	<20
Turbidity	NTU	<5
pH	-	6 – 9
Threshold odour number (TON)	TON	<100
BOD_5	mg/L	< 10
Ammonia nitrogen	mg/L	<1
Synthetic detergents	mg/L	<5

Note:

Apart from TRC which has been specified, the water quality standards for all parameters shall be applied at the point-of-use of the system. HU = Hazen Unit. NTU = Nephelometric Turbidity Unit.

The parameters are referenced to the target water quality standard adopted in the EIA Report for Hung Shui Kiu New Development Area Planning and Engineering Study

Sludge Handling and Disposal Routine Operation in YLS STW

7.6.4.7 The sewage sludge, being the by-product from the proposed YLS STW, will be undergone anaerobic digestion, followed by dewatering in the STW to form sludge cake. The sludge cake generated will be further transferred via sludge cake tankers to Sludge Treatment Facility (STF) at Nim Wan for further treatment. In order to minimise the public nuisance due to the routine transportation of sludge cake tankers from the STW to Nim Wan, only sludge cake tankers with fully enclosed container will be adopted for routine operations.

7.6.4.8 Apart from direct disposal in STF, other alternative will also be explored should the digestate be of suitable quality, including further treatment/ polishing/ sanitisation for conversion to other products like fertilisers or soil conditioner to support local agriculture/ horticulture industry.

Proposed Staging Plan of YLS Sewerage System

7.6.4.9 The YLS development will be developed into 4 stages. The proposed stage plan of YLS Sewerage System to cope with the staged intakes of YLS are elaborated below:

	Stage 1	Stage 2	Stage 3	Stage 4
Intake Year	2027-2029	2033	2038	2038
Total by Phase (m^3/day)	2,534	6,109	11,139	3,854
Cumulative Total (m^3/day)	2,534	8,643	19,782	23,635

7.6.4.10 The above detailed breakdown of sewage flows by different Stages are calculated and summarised in **Appendix 7.3**.

7.6.4.11 Stage 1 (Year 2027): it is proposed that the 1,800m^3/day sewage flow arising from YLS Stage 1 to be conveyed via the existing Lung Ting Pumping Station (LTPS) to SWSTW from year 2027 to 2029 as an interim arrangement. After the commencement of operation of YLS STW in 2030, the 1,800m^3/day of sewage generated in the YLS stage 1 will be diverted back to YLS STW for on-site treatment. In addition, the sewage from existing R5(EDA) will also be conveyed to YLS STW after commissioning in Year 2030.

The Stage 1 sewerage work arrangement as aforementioned has sought EPD no objection in principle. More details can be referred to **Appendix 7.4.**

7.6.4.12 Advance Stage 2 (Year 2030): YLS Stage 2 development will involve majority part of the proposed sewerage system including the SPS and on-site YLS STW as shown in **Figure 7.3**. The SPS and STW can be partially commission to cope with the flow in Stage 2 with the flexibility for further expansion to cope with the flow in subsequent stage.

7.6.4.13 Stage 3-4 (Year 2038): the proposed sewerage system for YLS Stage 3-4 developments will involve the construction of all proposed sewerage as shown in **Figure 7.3** and full commissioning of the proposed SPSs and on-site YLS STW.

7.6.5 Contingency plan for the proposed SPS(s) and YLS STW

7.6.5.1 Given the sensitivity of inner Deep Bay in term of water quality and ecology, extensive effort will be expedited to avoid the occurrence for emergency discharges. In order to achieve this, the design of SPSs and STW will be cautiously reviewed to include additional provisions including as follows:

- Twin rising mains would be provided. Should one of the duty mains be taken out of operation, the remaining one would still be able to deliver flow;
- Standby pumps and treatment facilities would be provided in case of unexpected breakdown of pumping and treatment facilities such that the standby pumps and treatment facilities could take over and function to replace the broken pumps and treatment facilities; and
- Dual electricity supply or backup power supply facilities such as diesel generator would be provided in case of power failure to sustain the function of pumping and treatment facilities.

7.6.5.2 Under very unlikely extreme scenario, YLS STW and SPSs will still have emergency discharge to the downstream water receiving body as described below. Nevertheless, in order to minimise the potential water quality impact in case of emergency discharge, the future operator will be required to develop a Contingency Plan in the detailed design stage to deal with the remote occurrence of emergency discharge. The Contingency Plan should include the following items.

- Locations of water bodies or WSRs in the vicinity of the emergency discharges;
- A list of relevant government departments to be informed and to provide assistance in the event of emergency discharge, including key contact persons and telephone numbers;
- Reporting procedures required in the event of emergency discharges; and
- Procedures listing the most effective means in rectifying the breakdown of the SPS in order to minimise the discharge duration.

Emergency Discharge Scenarios for the New YLS STW and SPSs		
Scenario	Discharge Route	Receiving Water Body
YLS STW not functioning	All Raw Sewage -> Reedbed -> Deep Bay	Deep Bay

SPSs not functioning	All Raw Sewage -> nullahs near Shan Ha Road and Kiu Hing Road -> Deep Bay	Deep Bay
The NWNT effluent tunnel / the Urmston Road submarine outfall not in operation (if surplus TSE would discharge to the inlet chamber of the NWNT effluent tunnel at San Wai for discharging to the Urmston Road submarine outfall)	Surplus TSE -> Reedbed -> Deep Bay	Deep Bay

7.7 Conclusions and Recommendations

7.7.1.1 This chapter has reviewed the existing and planned sewerage infrastructure within and in the vicinity of the Study Area. This chapter also collected information of sewage collection, treatment and disposal arrangements in the YLSTW and SWSTW sewerage catchments.

7.7.1.2 The sewage flows and loads generation from the YLS PDA are estimated and the proposed sewage treatment and disposal strategy for YLS are accordingly formulated, which involve discharging the YLS sewage flows by the proposed sewerage system to a new on-site STW in the southern part of the PDA for treatment, followed by the reuse of TSE as reclaimed water as non-potable uses in PDA and adjacent developments.

7.7.1.3 Equivalent pollution loading from tertiary TSE in YLS STW (upon offsetting by credit gained from the existing livestock farms) could be discharged to Yuen Long Nullah adjacent to the STW with strict compliance to the requirement for "No Net Increase in Pollution Loading", while reedbed is provided to further polish a small amount of TSE that discharge to Yuen Long Nullah and Deep Bay. The rest of sewage effluent will be treated by secondary plus treatment and discharge to the Urmston Road submarine outfall.

7.7.1.4 It should also be noted that the STW and the reclaimed water reuse activity are separate DPs to be constructed and operated by the future operators. Separate EIA shall be prepared and submitted to EPD for approval under the EIA Ordinance to apply for the Environmental Permit.